Passages - Seven and a Half Lessons About the Brain

When it comes to body budgeting, prediction beats re-action.
Why did a brain like yours evolve? That question is not answerable because evolution does not act with purpose there is no "why”. But we can say what is your brain's most important job.
It's not rationality. Not emotion. Not imagination, or creativity, or empathy. Your brain's most important job is to control your body - to manage allostasis - by predicting energy needs before they arise so you can efficiently make worthwhile movements and survive. Your brain continually invests your energy in the hopes of earning a good return, such as food, shelter, affection, or physical protection, so you can perform nature's most vital task: passing your genes to the next generation.

Your brain network is organized in much the same an Its neurons are grouped into clusters that are like airports.
Most of the connections in and out of a cluster are local so, like an airport, the cluster serves mostly local traffic. In addition, some clusters serve as hubs for communication. They are densely connected to many other clusters and some of their axons reach far across the brain and act as long-distance connections. Brain hubs, like airport hubs, make a complicated system efficient. They allow most neurons to participate globally even as they focus more locally. Hubs form the backbone of communication throughout the brain.
Hubs are supercritical infrastructure.
world. Hub damage is associated with

depression, schizo-phrenia, dyslexia, chronic pain, dementia, Parkinson's disease, and other disorders. Hubs are points of vulnerability because they are points of efficiency - they make it possible to run a human brain in a human body without depleting a body budget.

Our nervous systems are tended by others. Your brain secretly works with other brains. This hidden cooperation keeps us healthy, so it matters how we treat one another in a very real, brain-wiring way. Therefore, we're not only more responsible for babies and for ourselves than we might think; we're also more responsible for other adults than we might think. Or want. Like it or not, we influence the brains and bodies of those around us with our actions and words, and they return the favor.

Mankind has a single brain architecture a complex network and yet each individual brain tunes and prunes itself to its surrounding. The mind and the body are strongly linked, and the boundary between the two is porous. Your brain's predictions prepare your body for action and then contribute to what you sense and otherwise experience.